
Introduction to Machine Learning

*From Basic Concepts to
Decision Trees
& Random Forests*



Can a Computer Learn This?

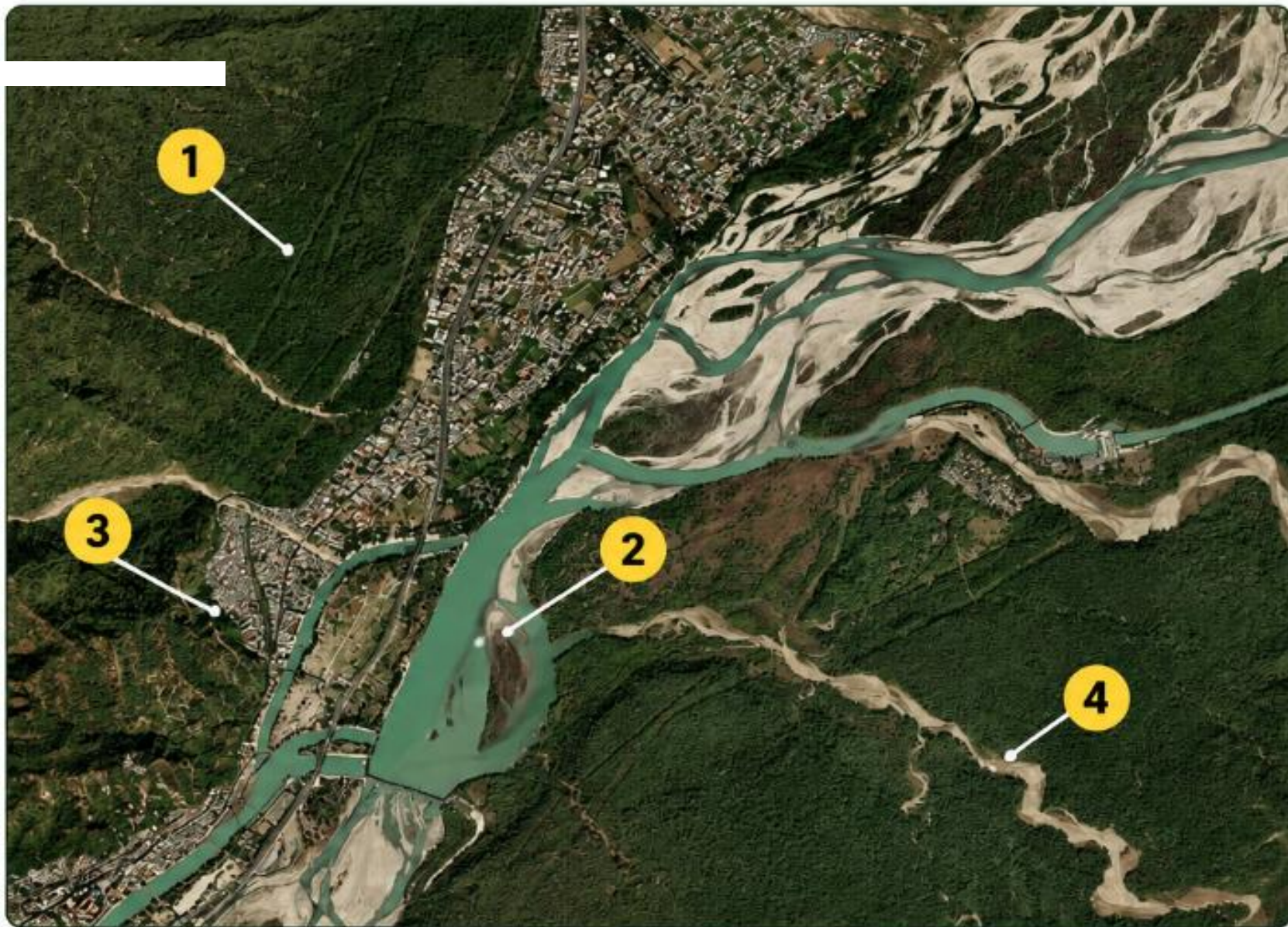
*Look at this satellite image.
What do you see?*



Think for a moment!

What do you think the
numbered areas represent?

- 1 What is area 1?
- 2 What is area 2?
- 3 What is area 3?
- 4 What is area 4?



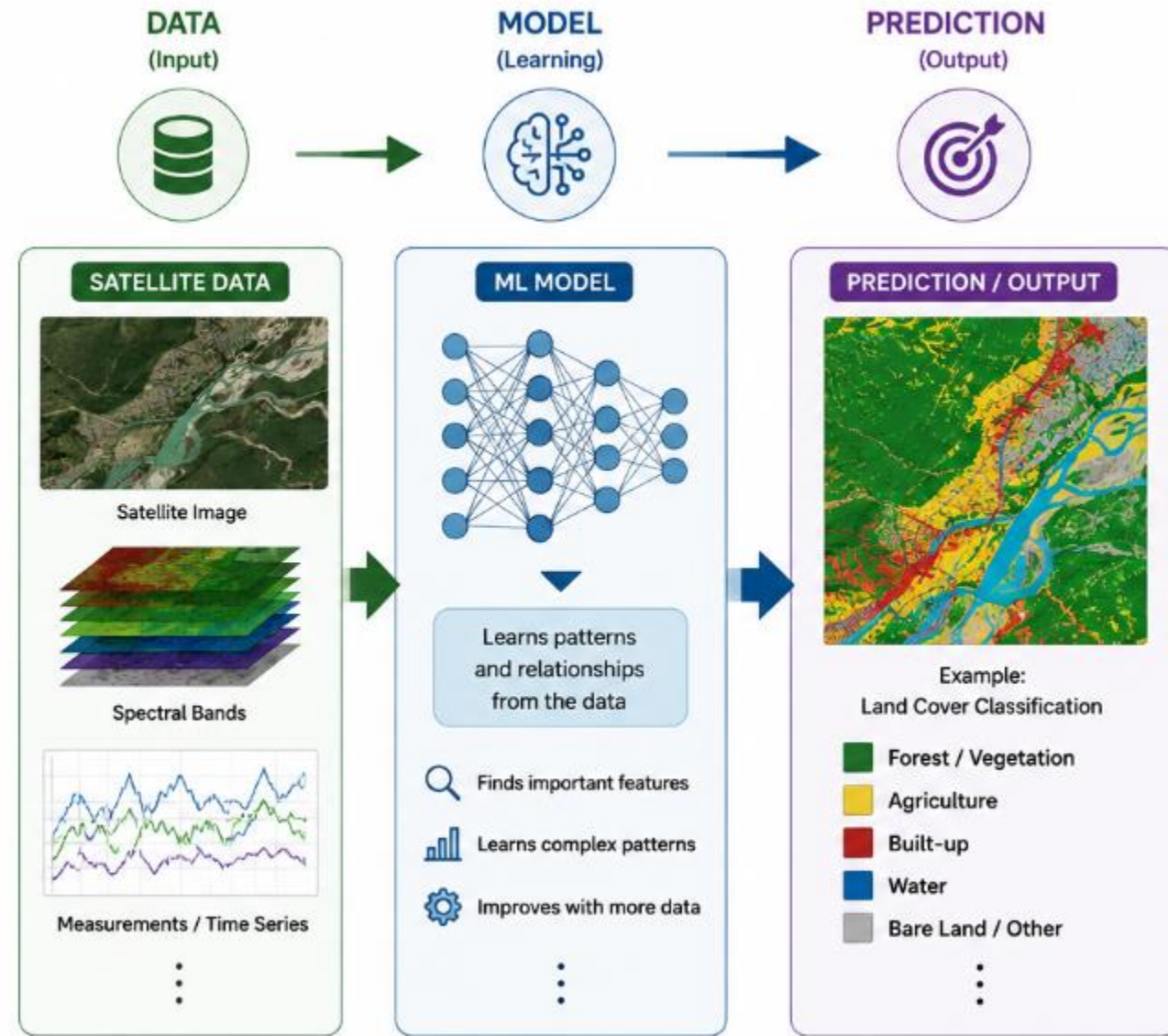
**Can we teach a computer to identify
these features automatically?**



**This is where
Machine Learning
comes in.**

What is Machine Learning?

- **Main Point:** Teaching computers to learn patterns from data and use those patterns to make predictions or decisions.
- **Key Concepts:**
 - **Data:** The fuel for learning (numbers, text, images).
 - **Model:** The mathematical engine that finds patterns.
 - **Prediction:** Using the model to make decisions on new, unseen data.



Machine Learning =
Learning patterns from data → Applying them to new data



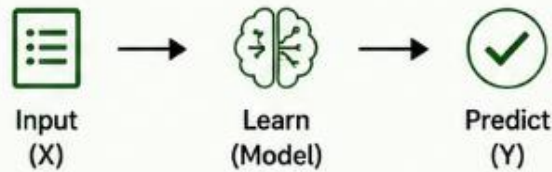
Once trained, the model can make predictions on **new, unseen data** quickly and consistently.



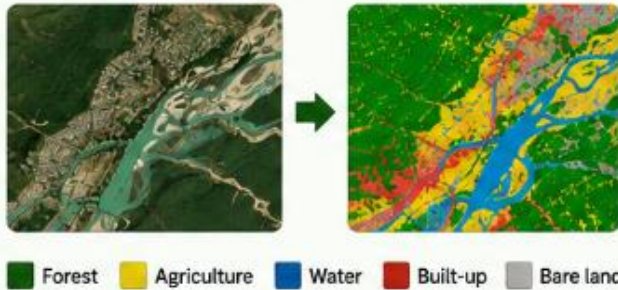
Types of Machine Learning

1 SUPERVISED LEARNING

Learn from labelled examples



EXAMPLE Satellite image → Land-cover class



★ We provide the correct answers.
The model learns to map inputs to outputs.

2 UNSUPERVISED LEARNING

Find patterns without labels



EXAMPLE Satellite pixels → Natural clusters



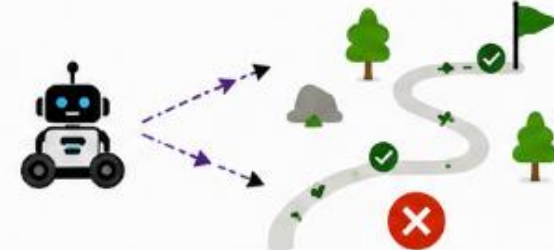
★ No correct answers are given.
The model discovers patterns by itself.

3 REINFORCEMENT LEARNING

Learn through rewards and penalties



EXAMPLE Agent learns to navigate



★ The agent tries actions, receives feedback,
and improves over time.



TODAY:
SUPERVISED LEARNING

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Decision Trees

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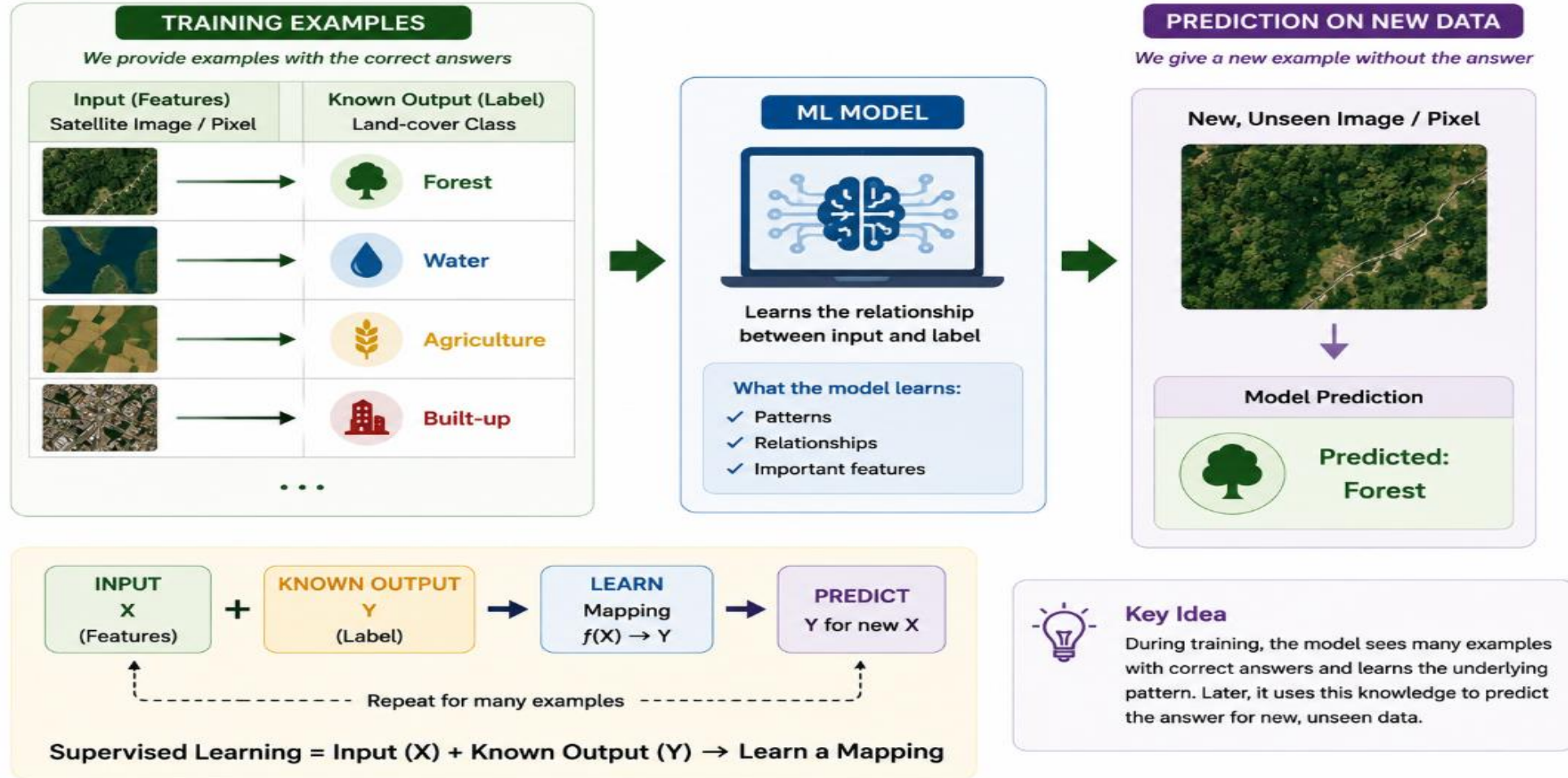


Random Forests

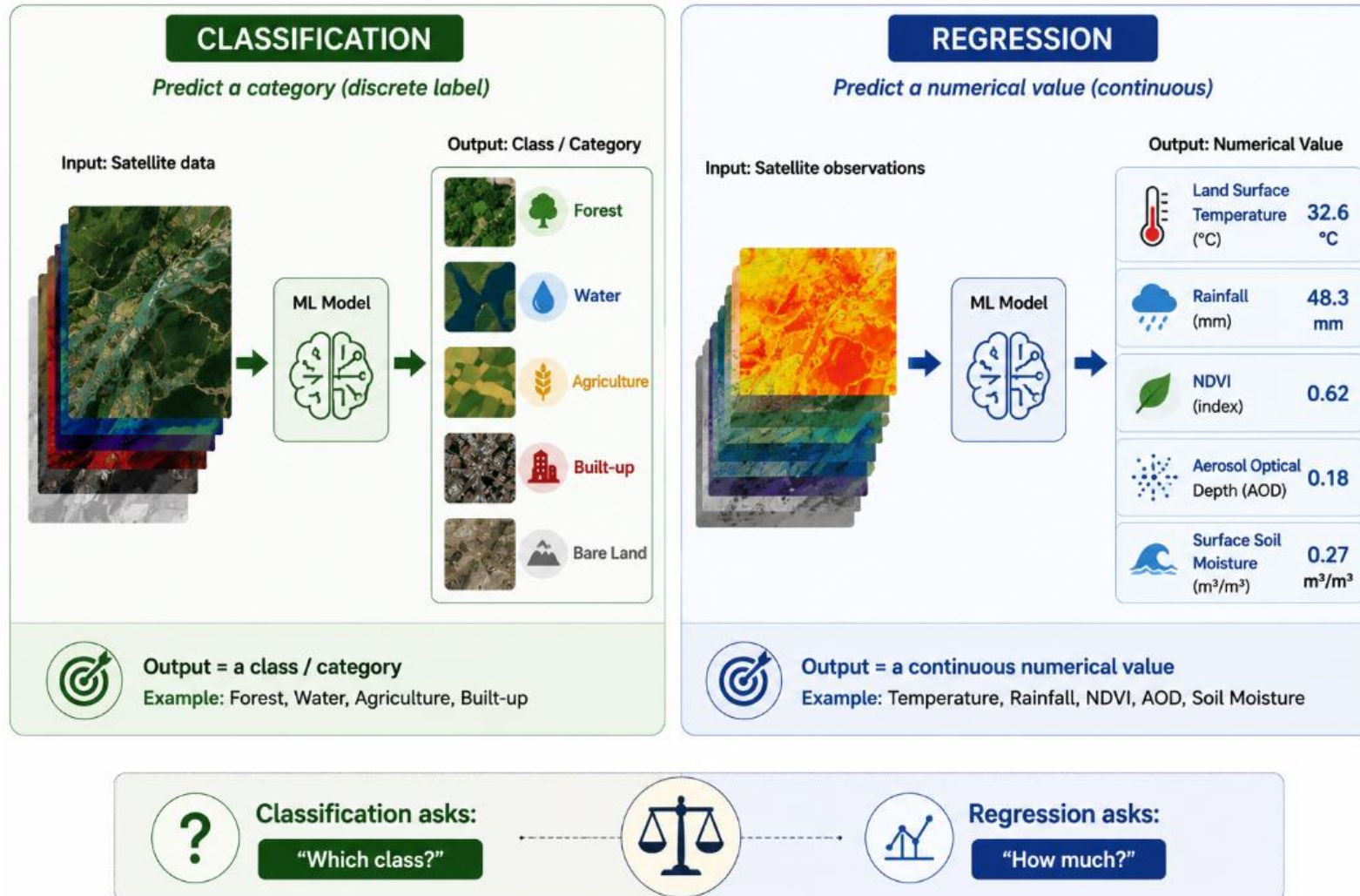
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...
And more

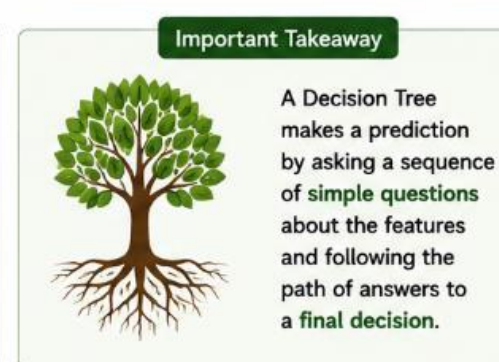
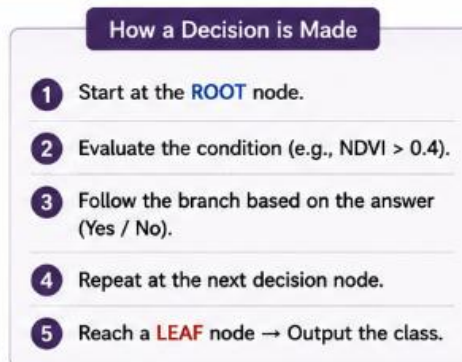
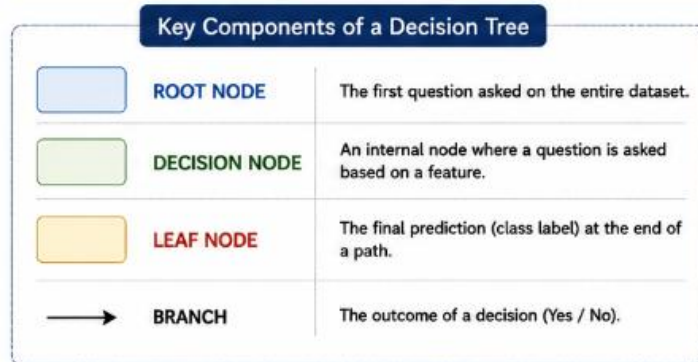
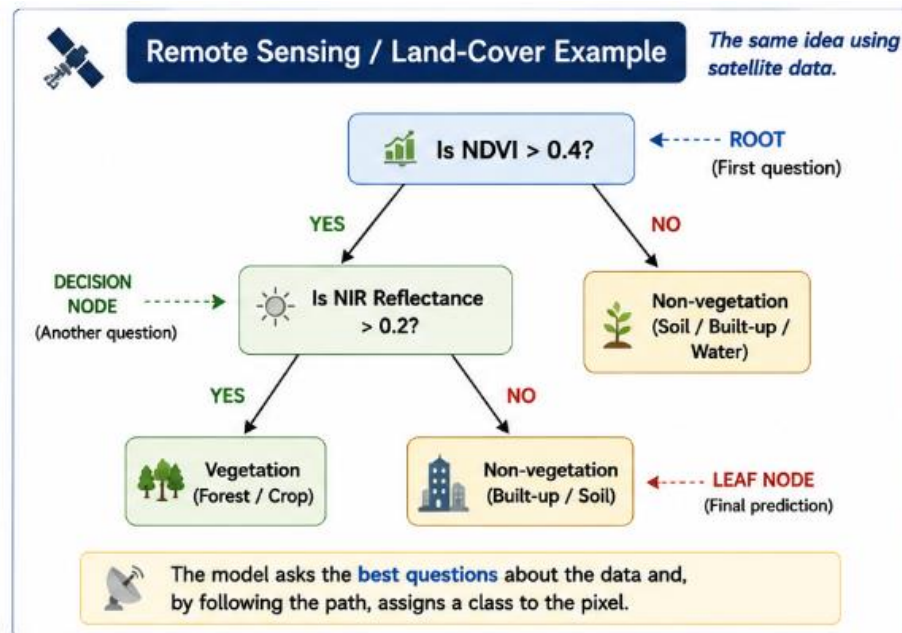
Supervised Learning : Learning from Examples



What Are We Trying to Predict?

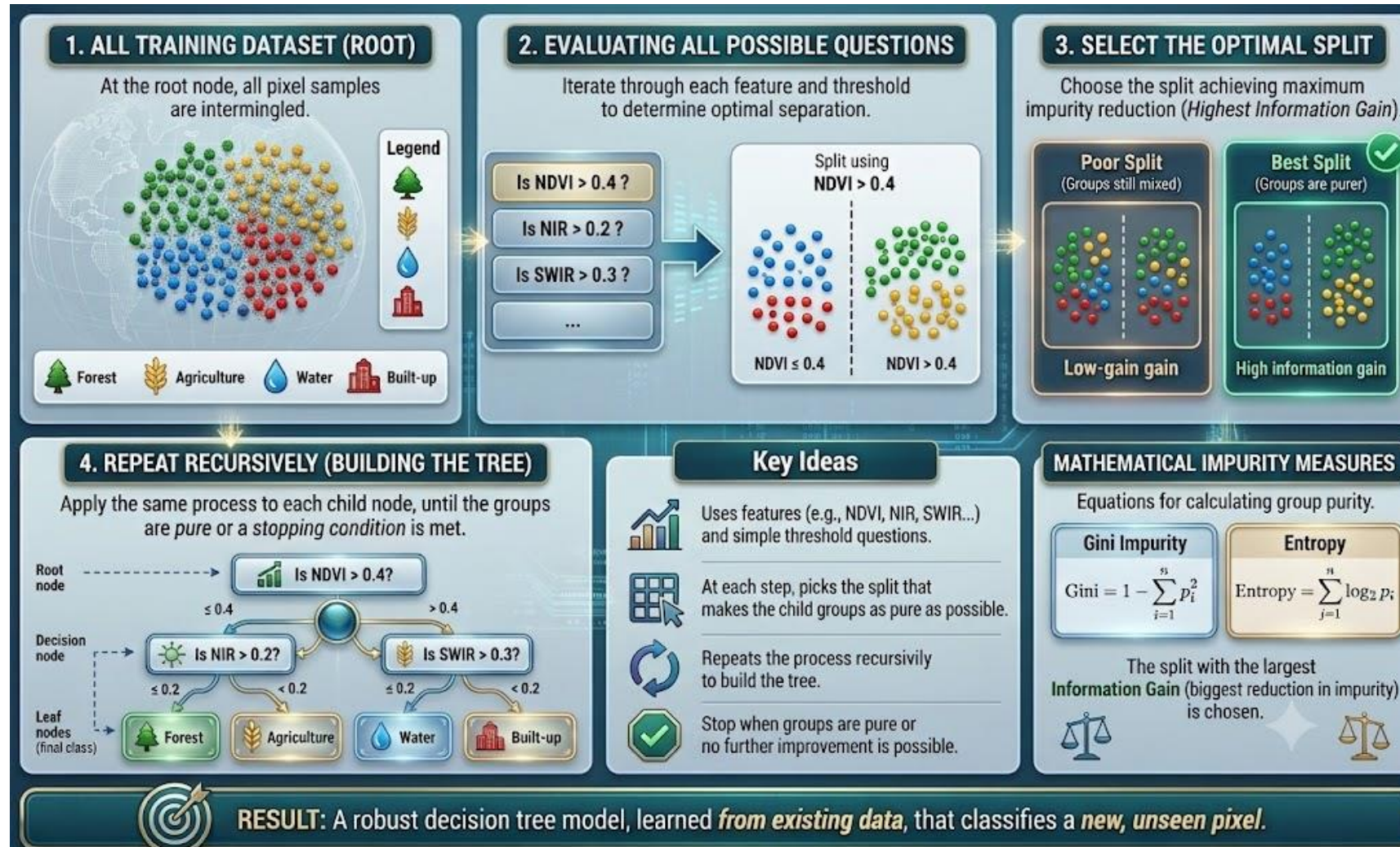


Decision Tree: A Sequence of Simple Decisions

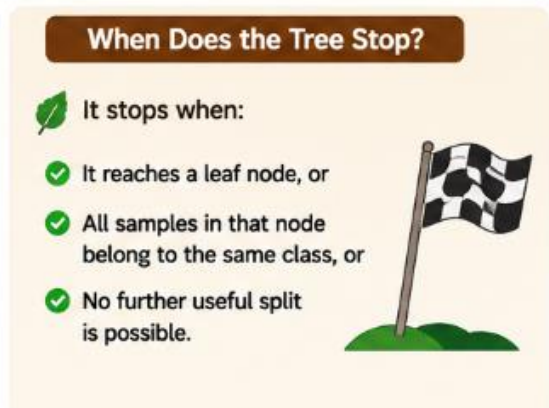
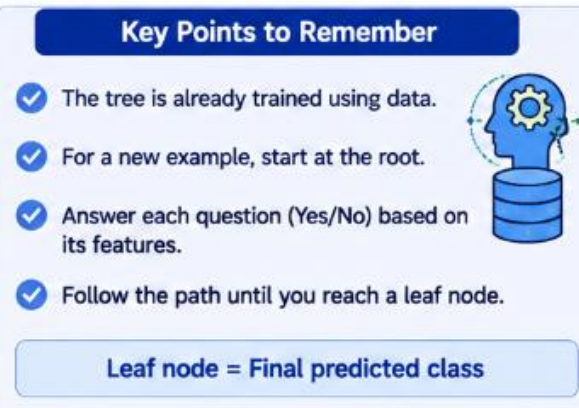
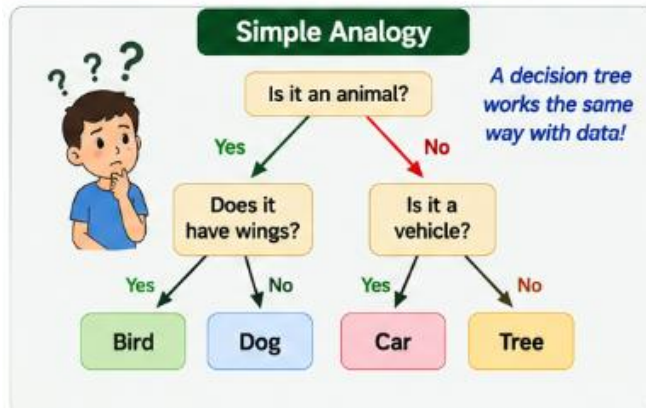
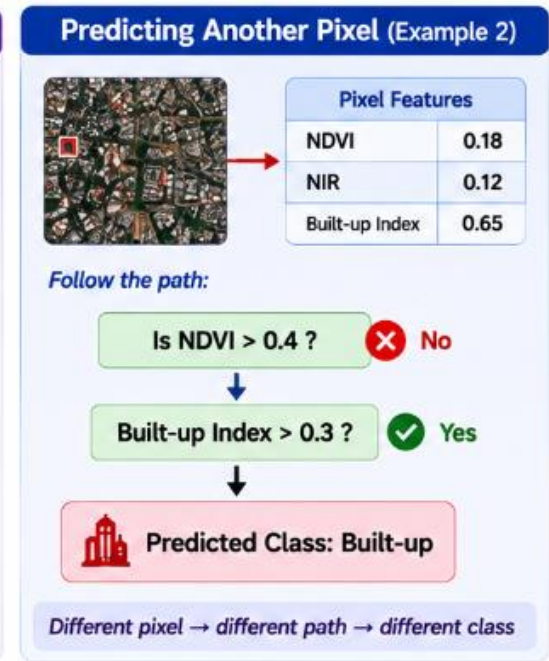
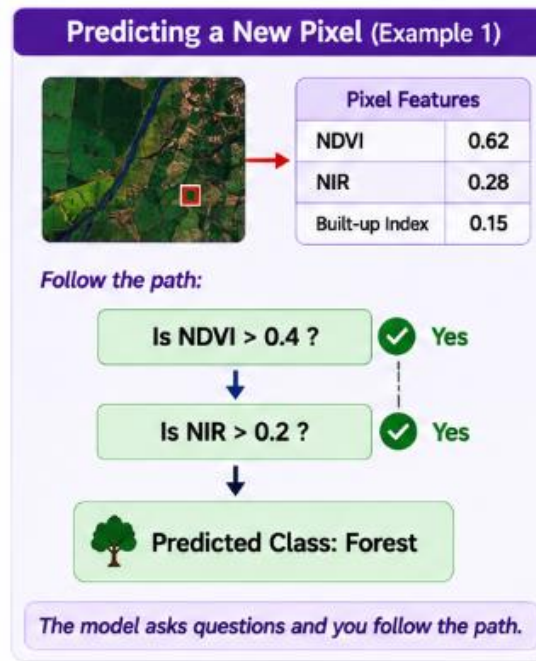
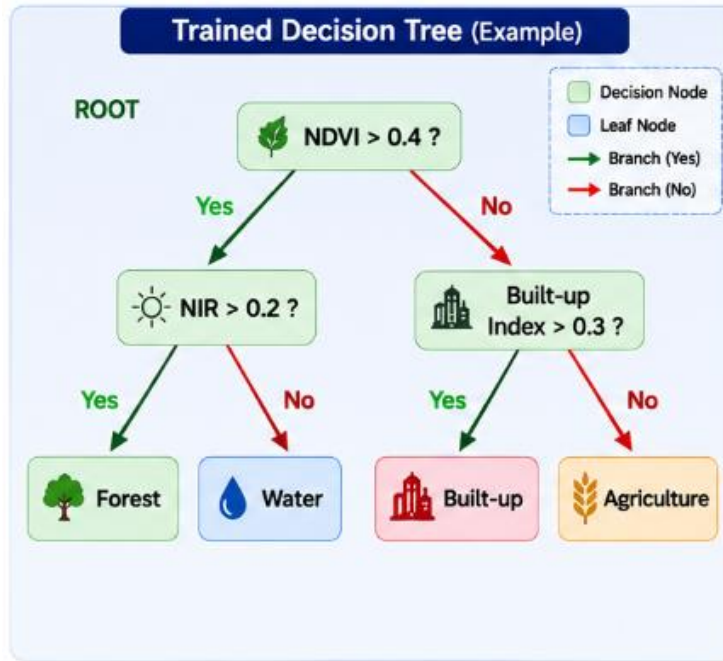


Decision Tree = A sequence of **simple decisions** (Yes / No questions) leading to a **final prediction** (a class).

How Decision Tree Learn



Using a Decision Tree

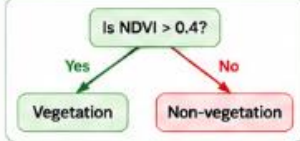


Pros & Cons of Decision Trees

✓ PROS (Advantages)



1. Easy to Understand
The tree structure is intuitive and easy to interpret.



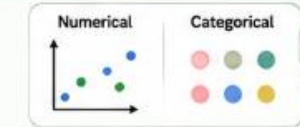


2. Interpretable
We can see which features are used and how decisions are made.



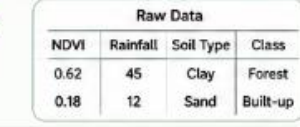


3. Works with Any Data Type
Handles both numerical and categorical features.





4. Requires Little Data Preparation
No need for scaling or normalization. Less preprocessing needed.



NDVI	Rainfall	Soil Type	Class
0.62	45	Clay	Forest
0.18	12	Sand	Built-up

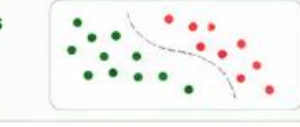


5. Fast to Train and Predict
Training and prediction are computationally efficient.





6. Handles Non-linear Relationships
Can capture complex patterns through hierarchical splits.



✗ CONS (Disadvantages)

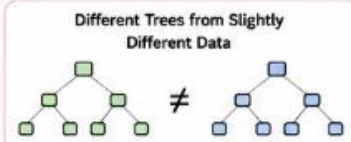


1. Prone to Overfitting
If the tree is too deep, it learns noise in the training data.



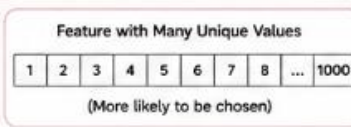


2. High Variance (Unstable)
Small changes in data can result in a very different tree.





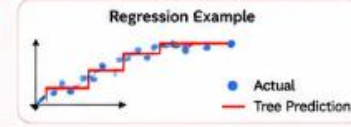
3. Biased Towards Features with Many Levels
May prefer features with many unique values.



1	2	3	4	5	6	7	8	...	1000
(More likely to be chosen)									



4. Piecewise Constant Predictions
Predictions are in "steps" and may not be smooth.



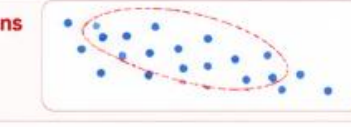


5. Single Tree Has Limited Power
A single tree may not perform well on complex problems.





6. Harder to Capture Global Patterns
Focuses on local splits and may miss global structure in data.



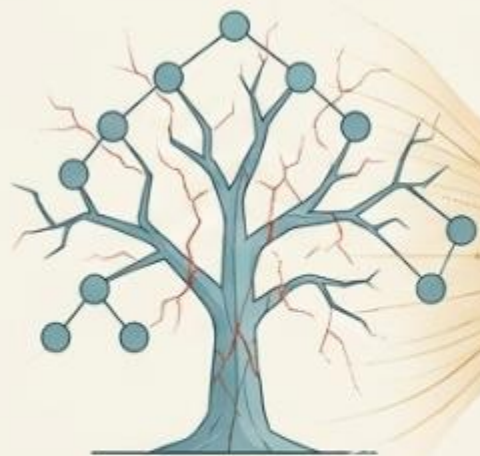


Key Takeaway: Decision Trees are easy to interpret and fast, but can overfit and be unstable. That's why ensemble methods like **Random Forests** combine many trees to get the best of both worlds!



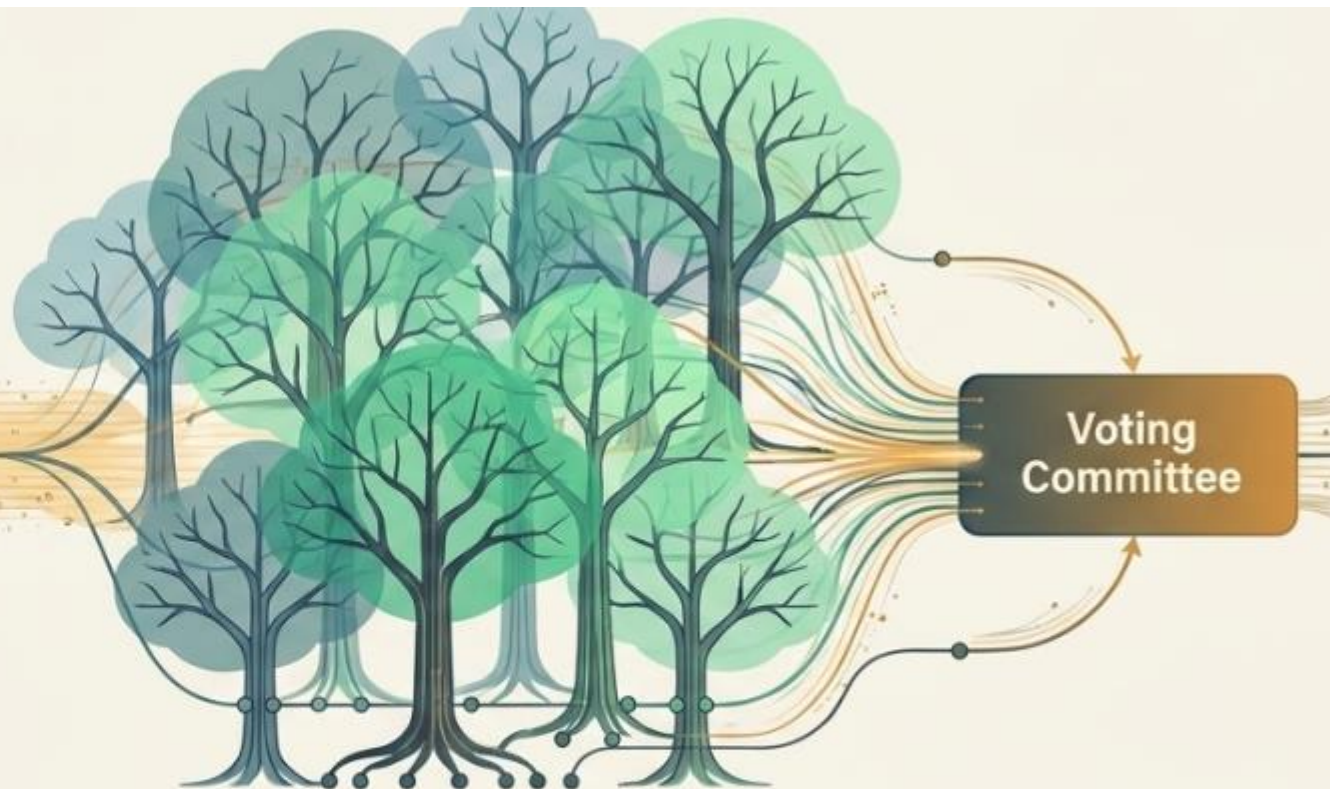
Paradigm Shift: Ensemble Learning

Solving mathematical instability
with the Wisdom of the Crowd.



Single Decision Tree:
Prone to Overfitting

A **single decision tree** is like relying on the opinion of one highly opinionated expert—prone to bias and likely to memorise specific case studies rather than universal rules.



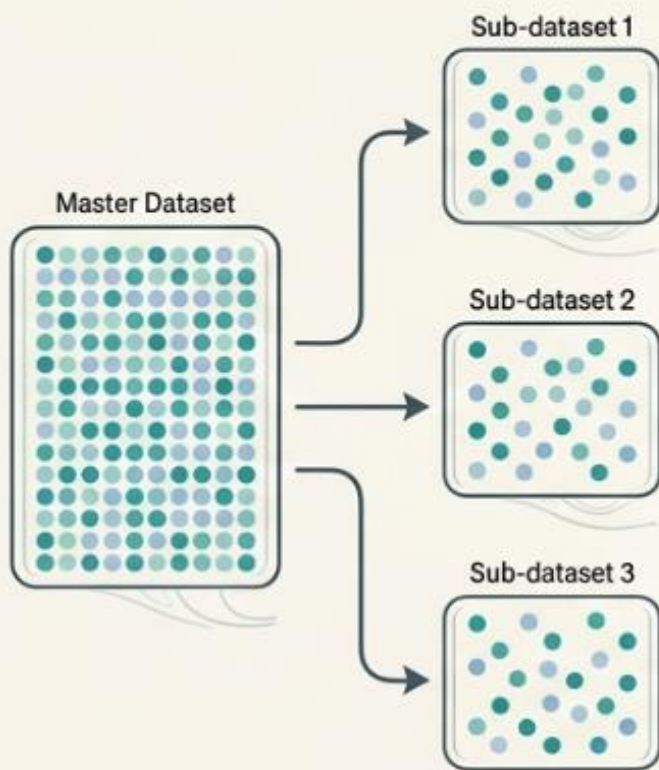
The Random Forest Solution:

Instead of building one massive, overfitted model, we train hundreds of distinct, uncorrelated decision trees. By combining their individual perspectives, the final ensemble model cancels out individual errors, resulting in a highly accurate, robust system that dramatically outperforms any single tree.

Random Forest

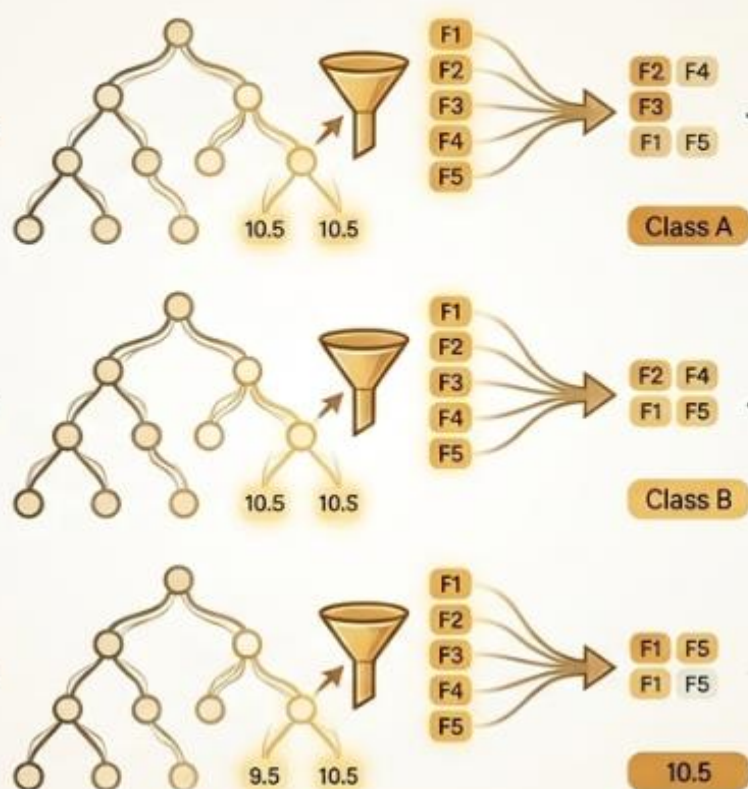
Step 1: Bootstrapping (Data Diversity)

The algorithm creates multiple different sub-datasets by randomly sampling the original data with replacement. Every tree gets a slightly different perspective of the world.



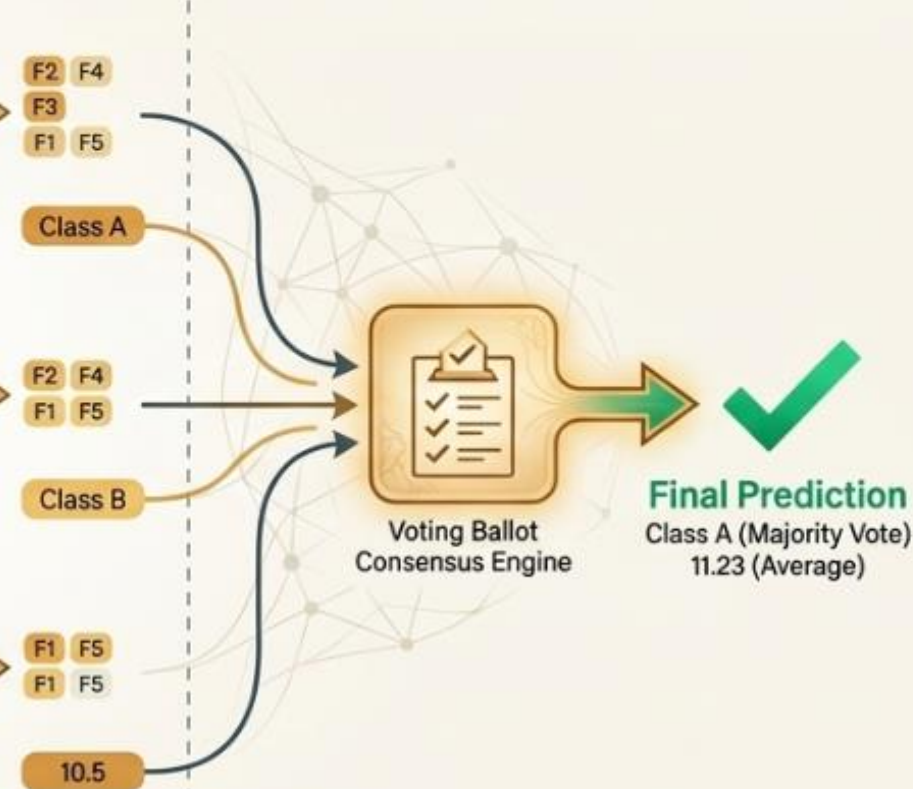
Step 2: Feature Randomness (Structural Diversity)

To ensure the trees do not all make the exact same decisions, they are forced to be structurally blind. At every single split, the tree is only allowed to consider a random subset of features.



Step 3: Aggregation (The Final Consensus)

The individual, independently trained trees are brought together to evaluate new data. The forest produces a single, highly accurate prediction through collective consensus.



Random Forest vs Decision Tree

Single Decision Tree	Random Forest Ensemble
☐ Architecture: One deeply grown model.	☒ Architecture: Hundreds of uncorrelated models acting together.
☒ Interpretability: Extremely high. Logic can be mapped visually.	☐ Interpretability: Low (a "Black Box" model). Difficult to trace exactly why a specific decision was made.
☐ Performance: Brittle. Highly prone to overfitting and memorising noise.	☒ Performance: Highly robust. Immune to minor variations in training data.
☒ Speed: Near-instantaneous training and prediction.	☐ Speed: Computationally heavy; requires more time and resources.
☒ Best For: Simple datasets where explaining the decision is more important than absolute accuracy.	☒ Best For: Complex, high-stakes real-world environments where precision and reliability are non-negotiable.

Applications in Remote Sensing



Thank You